

The background is a deep blue gradient with a starry space texture. Overlaid on this are several faint, light-colored circular patterns. Some are solid lines, while others are dashed. A prominent circular scale with numerical markings (140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260) is visible on the left side. Other circular elements include concentric circles and arcs, some with arrows indicating a clockwise direction.

LECTURE NO (8)

TESTING WEB APP AND MAINTENANCE



Testing web applications

Terminology

- **Some definitions:**
 - **Testing:** An activity conducted to evaluate the quality of a product to improve it by identifying defects and problems.
 - **Error:** the actual result deviates from the expected.
 - Our expected results should (theoretically) come from our requirements definition.
 - Most often, the goals/concerns/expectations of stakeholders serve as the testing basis.
 - **Test case:** a set of inputs, execution conditions, and expected results for testing an object.

Test Objectives

- Main objective: find errors, NOT show that none exist.
- Complete test coverage is impossible, so testing focuses on mitigating the largest risks.
 - Where's the greatest potential for loss?
 - What are the sources of this risk?
- Start testing as early as possible – even with restricted resources and time.

Test Levels

- ❑ **Unit tests:** test the smallest testable units (classes, Web pages, etc.), independently of one another. Unit testing is done by the developer during implementation.
- ❑ **Integration tests:** evaluate the interaction between distinct and separately tested units once they have been integrated. Integration tests are performed by a tester, a developer, or both jointly.
- ❑ **System tests:** test the complete, integrated system. System tests are typically performed by a specialized test team.

Test Levels (Cont..)

❑ Acceptance tests: evaluate the system in cooperation with or under the auspice of the client in an environment that comes closest to the production environment. Acceptance tests use real conditions and real data.

❑ Beta tests: let friendly users work with early versions of a product with the goal to provide early feedback. Beta tests are informal tests (without test plans and test cases) which rely on the number and creativity of potential users.

The Tester Role

- The ideal tester has a “destructive” attitude.
- Very difficult for developers to “destroy” their own work.
- However, Web projects focus heavily on unit tests, making them more prone to errors.
- Thus, some guidelines:
 - Have others in the Web team perform tests.
 - Best tester is the one who gets the most bugs *fixed*.

Test Methods and Techniques

- **Link Testing :** Finding broken links, orphan pages etc.
- **Browser Testing :** variety of browsers exists. Thus need to test.
- **Load Testing:** Does the system meet required response times and throughput?
- **Stress Testing:** How does the system behave under abnormal/extreme conditions?
- **Continuous Testing :** Typically, running the operation a few times doesn't produce an error, hence the need for continuous testing.
- **Security Testing:**
 - Is our SSL certificate working?
 - What happens if I try to access a protected page/site in a non-secure way (i.e., http://)?



THE IMPORTANCE OF **WEBSITE MAINTENANCE SERVICES**

In today's fast-paced digital world, having a well-designed website is just the beginning of your online journey. To ensure your website continues to perform optimally, regular maintenance is crucial.

1. ENHANCING SECURITY

One of the primary reasons to invest in Website Maintenance Services is to protect your site from security threats. Cyber attacks and hacking attempts are on the rise, and an outdated website is a prime target. Regular maintenance includes installing security patches, updating software, and monitoring for any suspicious activity.



2. BOOSTING PERFORMANCE

A slow and unresponsive website can drive visitors away. Website Maintenance Services ensure that your site runs efficiently by optimizing images, updating plugins, and removing unnecessary files. Regular performance checks and optimizations help in providing a seamless user experience, which is critical for retaining visitors and reducing bounce rates.





4. ENSURING COMPATIBILITY

Technology is constantly evolving, and new browsers, devices, and operating systems are regularly released. Website Maintenance Services help in ensuring that your website remains compatible with the latest technology.

3. IMPROVING SEO

Search engines favor websites that are well-maintained and regularly updated. Website Maintenance Services include tasks like fixing broken links, updating content, and ensuring that your site is mobile-friendly.



5. CONTENT UPDATES

Fresh and relevant content is key to engaging your audience. Website Maintenance Services often include content updates, ensuring that your website reflects the latest information about your products, services, and industry trends. Regularly updated content not only keeps your audience informed but also signals to search engines that your site is active and valuable.



6. BACKUPS AND RECOVERY

Data loss can occur due to various reasons, including cyber-attacks, server failures, or human error. Regular backups are an integral part of Website Maintenance Services. By maintaining up-to-date backups, you can quickly recover your site in case of data loss, minimizing downtime and ensuring business continuity.





7. ANALYTICS AND REPORTING



Understanding how your website is performing is crucial for making informed decisions. Website Maintenance Services often include monitoring website analytics and generating reports. These insights help you understand user behavior, track key performance metrics, and identify areas for improvement.

CONCLUSION

Investing in Website Maintenance Services is not just about keeping your site up and running; it's about ensuring that your website continues to support your business goals. From enhancing security and performance to improving SEO and user experience, regular maintenance is crucial for the long-term success of your website.



THE END